

Genetic Diversity Analysis in Homologous Genes of Mammals and Humans

Using *ALDH2* and *TAS2R38* Genes as Examples

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1 Introduction

2 Methods

3 Results

4 Discussion

Genetic Diversity: The Foundation of Evolution

The Significance of Genetic Variation

- Genetic diversity is the raw material for **evolution and adaptation**
- Single nucleotide polymorphisms (SNPs) account for most genetic variations in populations
- Understanding genetic diversity helps elucidate:
 - Evolutionary history and phylogenetic relationships
 - Population migrations and demographic events
 - Disease susceptibility and pharmacogenetics

Model Genes for Diversity Studies

- **ALDH2**: Aldehyde dehydrogenase, crucial in alcohol metabolism
 - rs671 (Glu504Lys) shows strong East Asian-specific distribution
 - Associated with alcohol flush reaction and disease risks
- **TAS2R38**: Bitter taste receptor for PTC/PROP compounds
 - Classic example of balanced polymorphism in human populations
 - Three linked SNPs determine taste sensitivity phenotypes

Research Objectives and Approach

Methodological Framework

Experimental Approach:

- DNA extraction from buccal swabs
- PCR amplification of target regions
- Sanger sequencing for genotype determination

Bioinformatic Analysis:

- Homologous sequence retrieval
- Multiple sequence alignment
- Phylogenetic tree construction

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Phenotypes Survey and Genes Extraction

Phenotype Assessment

- **Taste sensitivity:** PTC/PROP test strips for bitter taste perception
- **Aldehyde metabolism:** Recorded based on personal experience

Genomic DNA Extraction

Sample Collection:

- Buccal swabbing
- Two samples per person
- Air-dried in envelopes

Extraction Protocol:

- Proteinase K digestion
- DNeasy Mini Spin Column purification
- Elution in Buffer AE (50 μ L)

Quality Control

- **Spectrophotometry:** NanoDrop concentration measurement
- **Electrophoresis:** 1% agarose gel, 120 V, 1 kbp ladder
- **Storage:** Diluted to 5 ng/ μ L in Buffer AE

PCR Amplification

Primer Design and Preparation

- **Target genes:** *ALDH2* and *TAS2R38*
- **Mutation sites:**
 - *ALDH2*: Position 1,510
 - *TAS2R38*: Positions 145, 785, 886
- **Primer design:** Using Benchling Primer3 tool + universal M13 tails
- **Amplicon size:** 200 ~ 500 bp

PCR Amplification

Reaction Setup:

- 15 μ L volume each (Primer + Template + dNTP + Mg^{2+} + TaqGold + Solvent)
- 2 templates $\times$ 3 primers $\times$ 2 duplicates = 12 samples
- Blank control (water as template)

Thermal Cycling:

- Touchdown protocol
- 60 $\rightarrow$ 50 $^{\circ}$ C annealing (–2 $^{\circ}$ C every 2 cycles) & 42 final cycles
- 2.5 ~ 3 hours total

Sequencing and Phylogenetic Study

Analysis and Sequencing

- **Electrophoresis:** 1.5% agarose gel, 120 V, 100 bp Ladder
- **Sequencing:** 6 samples (clear bands & correct length) sent for sequencing
- **Analysis:** Retrieve sequence and identify genotypes of each sample

Phylogenetic Tree Construction

- **Homologous sequences:** Downloaded *ALDH2* & *TAS2R38* CDS from UCSC/NCBI (≥ 10 species)
- **Phylogeny:** Construct phylogenetic using MEGA 12
- **Constructing Methods:** Neighbor-joining (NJ), Maximum Parsimony (MP), Maximum Likelihood (ML)

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Survey on Phenotypes

No	Name	Self-reported Alcohol Metabolism Ability	Bitter Taste Self-perception	Bitter Test Strip (1~5)
1	Le Zhang	Drinks occasionally, rarely flush; from Shaanxi; parents do not flush	No distinct sensation when eating cruciferous vegetables	Slightly bitter, 1~2
2	Jinlin Liu	Drinks occasionally, rarely flush; from Hunan; both parents flush easily	Enjoys eating cruciferous vegetables	Slightly bitter, 1
3	Xiangyu Han	No alcohol consumption; from Shandong; father flushes when drinking, mother does not drink	Enjoys eating cruciferous vegetables	Somewhat bitter, 2
4	Xinyu Jiang	No alcohol consumption; from Jiangxi; father flushes when drinking, mother does not drink	Enjoys eating cruciferous vegetables	Somewhat bitter, 1
5	Yizhan Feng	Moderate flushing when drinking; from Zhejiang; cannot recall if parents flush	Enjoys eating cruciferous vegetables	Somewhat bitter, 1

DNA Extraction

NanoDrop

- Sample A (Yizhan Feng): 70.9 ng/ μ L
- Sample B (Xiangyu Han): 39.8 ng/ μ L

Electrophoresis

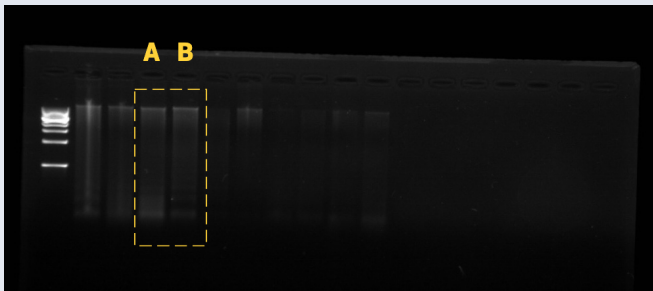


Figure 1: Electrophoresis of DNA stock

PCR Primer Design

Universal Primer

Universal primer M13 for best sequencing compatibility:

- F:5'-TGTAACACGACGGCCAGT-
- R:5'-CAGGAAACAGCTATGACC-

Gene	Position	Primer Sequence (5'→3')	T _m /°C	GC%	Size (bp)
<i>ALDH2</i>	1510	F: TGGAGCCCAGTCACCCTTTGGT R: atgggacagagcagaggctggg	61.8 62.1	59.1 63.6	413
<i>TAS2R38</i>	145	F: TTTCTGCACTGGGTGGCAACCA R: GCCGGCTGATGCTGAGACACAG	61.2 62.1	54.5 63.6	275
<i>TAS2R38</i>	785/886	F: TGCTCCTGCATCTGCACTGTCC R: CATGCCCAGAGGGACAAGCTGC	61.2 62.1	59.1 63.6	433

PCR Amplification

First PCR: Failure

- No distinct DNA bands in all sample lanes, only faint signals
- Presumably primer degradation

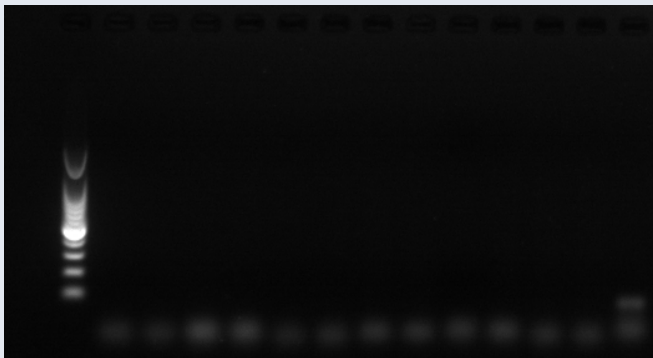


Figure 2: Leftmost: marker, rightmost: blank, samples are in between

PCR Amplification

Second PCR: Success

- Label:
Template[F(eng)/H(an)] + Primer[A(LDH2)/T(AS2R38)a/T(AS2R38)b]
- Distinct DNA bands in all sample lanes
- Two bands for Tb primer lane: PCR product (top) & primer dimer (bottom)

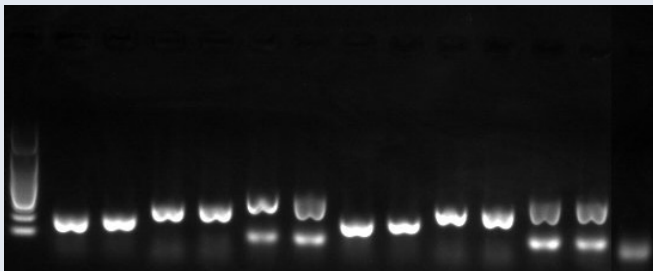


Figure 3: L → R: marker, FA×2, FTa×2, FTb×2, HA×2, HTa×2, HTb×2, blank

Sequencing Result

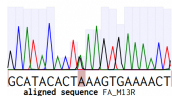
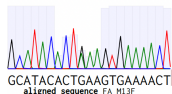


Figure 4: FA-1510

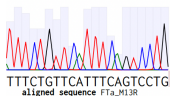
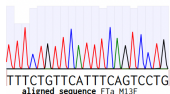


Figure 5: FTa-145

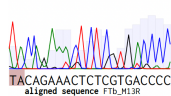
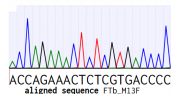
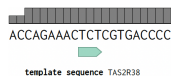


Figure 6: FTb-785

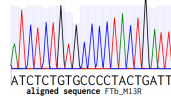
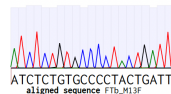
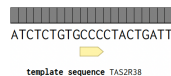


Figure 7: FTb-886

Sample A (Feng)

- G/A double peak at *ALDH2* c.1510 → **heterozygous G/A (rs671 GA)**
- Single A peak at *TAS2R38* c.145 → **wild type C/C**
- Single C peak at *TAS2R38* c.785 → **wild type C/C**
- Single C peak at *TAS2R38* c.886 → **wild type G/G**

Sequencing Result

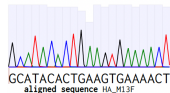


Figure 8: FA-1510

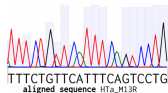
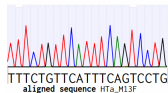


Figure 9: FTa-145

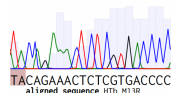
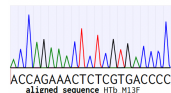


Figure 10: FTb-785

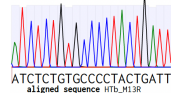
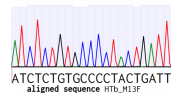


Figure 11: FTb-886

Sample B (Han)

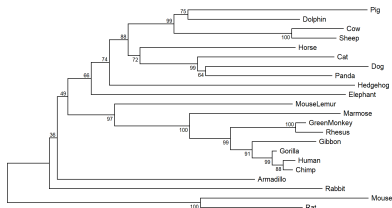
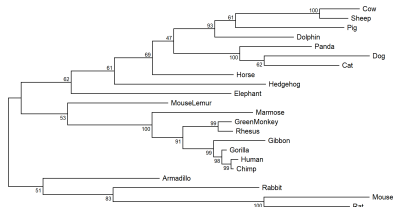
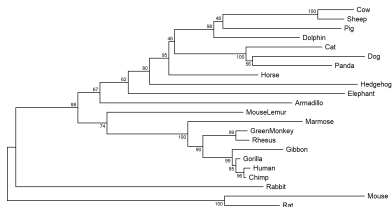
- Single G peak at *ALDH2* c.1510 → **wild type G/G**
- Single A peak at *TAS2R38* c.145 → **wild type C/C**
- Single C peak at *TAS2R38* c.785 → **wild type C/C**
- Single C peak at *TAS2R38* c.886 → **wild type G/G**

Phylogenetic Trees

Sources & Presets

- 22 species: Human, Chimp, Gorilla, Gibbon, Green Monkey, Rhesus, Marmoset, Mouse Lemur, Mouse, Rat, Rabbit, Hedgehog, Pig, Cow, Sheep, Dolphin, Horse, Dog, Panda, Cat, Elephant, Armadillo
- 3 methods: Neighbor Joining (NJ), Maximum Parsimony (MP), Maximum Likelihood (ML)
- 1000 times bootstrap tests
- Generate using MEGA 12

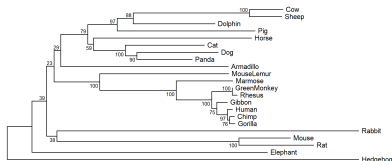
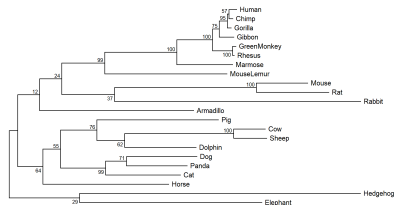
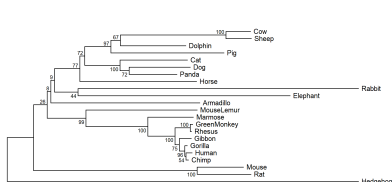
Phylogenetic Trees



Phylogeny of *ALDH2*

- Top-left: ML method
- Top-right: MP method
- Bottom-left: NJ method

Phylogenetic Trees



Phylogeny of *TAS2R38*

- Top-left: ML method
- Top-right: MP method
- Bottom-left: NJ method

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Phenotype-Genotype Correlation

Sample	Site	Genotype	Predicted Phenotype	Actual Phenotype
Han	<i>ALDH2</i> c.1510	Wild Type G/G	Normal aldehyde metabolism	No alcohol consumption
	<i>TAS2R38</i> c.145	Wild Type C/C	Strong bitter taste	Slightly bitter
	<i>TAS2R38</i> c.785	Wild Type C/C		
	<i>TAS2R38</i> c.886	Wild Type G/G		
Feng	<i>ALDH2</i> c.1510	Heterozygous G/A	Aldehyde metabolism disorder	Flush when drinking
	<i>TAS2R38</i> c.145	Wild Type C/C	Strong bitter taste	Slightly bitter
	<i>TAS2R38</i> c.785	Wild Type C/C		
	<i>TAS2R38</i> c.886	Wild Type G/G		

- Good correlation between genotypes and phenotypes for *ALDH2*
- Contradiction between genotypes and phenotypes for *TAS2R38*

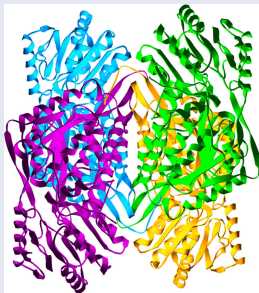
ALDH2 504Lys*

Overview

- Mitochondrial aldehyde dehydrogenase (*ALDH2*) is one of the most important enzymes in human alcohol metabolism.
- The *ALDH2*2 504Lys* variant (*rs671*) takes place at c.1510 (G → A) and substitutes **Glu** to **Lys** at residue 504.
- Inherited as an autosomal dominant trait:
 - Glu504 homozygotes (*ALDH2*1/1*) exhibit normal enzymatic activity
 - Glu504/Lys504 heterozygotes (*ALDH2*1/2*) show approximately 6% of normal activity ([Xiao et al., 1996](#))
 - Lys504 homozygotes (*ALDH2*2/2*) show negligible activity toward acetaldehyde ([Crabb et al., 1989](#))

ALDH2 504Lys*

Molecular Mechanism



Glu504 (negatively charged) to Lys (positively charged) substitution disrupts a critical salt bridge and hydrogen-bonding network at the tetramerization interface. This not only destabilizes the tetramer but also distorts the geometry of the catalytic and coenzyme(NAD^+)-binding sites in adjacent subunits. (Larson et al., 2005)

Figure 12: *ALDH2* tetramer

Larson et al. (2005) state that the mutation “results in a complete loss of activity”, and the dominant-negative effect occurs because “the incorporation of even a single subunit... is sufficient to destabilize the tetramer and inactivate the enzyme”.

ALDH2 504Lys*

Only one sample (Feng, fr. Zhejiang) among five harbored a mutation at *ALDH2* c.1510, characterized as a **G/A heterozygote**.

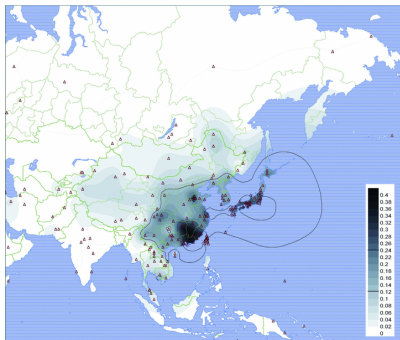


Figure 13: *ALDH2* 504Lys* distribution

Geo-Distribution

Li et al. (2009) reported the variant to be **essentially exclusive to East Asia**, and the highest frequency (up to 40.9%) was found in **Southeastern China**.

The expansion and genetic influence of the Han Chinese spread the allele throughout East Asia.

ALDH2 504Lys*

Evolution History

Li et al. (2009) concluded that allele was likely originated in ancient Han Chinese in Central China, although being found most abundant in Southeastern China.

- Indigenous populations in South China (e.g., Hmong-Mien, Daic groups like Kam and Laka) and the aboriginal peoples of Taiwan and Hainan have **very low or absent** frequencies of the allele.
- Historical and genetic data show these groups are descendants of migrants from Central China. The high frequency was **carried** and **maintained** by these migrants, not originated in the south.
- Historically, Altaic populations (e.g., Xianbei, Huns, Khitans, Jurchens) moved into Central China. Since modern Altaic populations have very low *ALDH2* 504Lys* frequencies, their integration into the Central Chinese gene pool would have diluted the allele's frequency in its place of origin.

ALDH2 504Lys*

Evolution History

- The Han Chinese migrants who moved south may have experienced:
 - Founder effect: The migrating groups carried a high frequency of the allele.
 - Protection against severe alcoholism: In ancient agrarian societies, even a modest reduction in alcohol consumption could have conferred significant survival and reproductive advantages by preventing liver disease, accidents, and social strife.

Associated Diseases

- Cardiovascular Disease: *ALDH2* is the primary enzyme responsible for converting nitroglycerin to nitric oxide. *ALDH2* 504Lys* carriers show a significantly blunted response to nitroglycerin, a major issue in cardiovascular pharmacogenetics. ([Li et al., 2006](#))
- Cancer Relations: Cancer incidences increase among alcoholics in organs including esophagus, stomach, liver, upper aerodigestive tract in which acetaldehyde is produced by the alcohol dehydrogenases. ([Yokoyama, 2001](#))

TAS2R38 SNPs

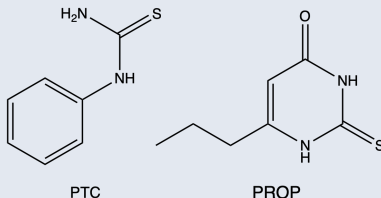
Overview

- TAS2R38 encodes a G protein-coupled bitter taste receptor protein found on the plasma membrane of taste cells, which determines an individual's ability to taste certain bitter compounds like 6-n-propylthiouracil (PROP) and phenylthiocarbamide (PTC).
- Contain 3 common SNPs:
 - *rs713598*: Pro → Ala at 49;
 - *rs1726866*: Ala → Val at 262;
 - *rs10246939*: Val → Ile at 296.
- Strong linkage disequilibrium ([Risso et al., 2016](#)), forming 2 common observable haplotypes: PAV (the “taster” allele) and AVI (the “non-taster” allele).

TAS2R38 SNPs

Molecular Mechanism

- *TAS2R38* encodes a Class A GPCR expressed in type II taste receptor cells of the tongue.
- *TAS2R38* responds to compounds containing **N-C=S moiety**. (Bufer et al., 2005)

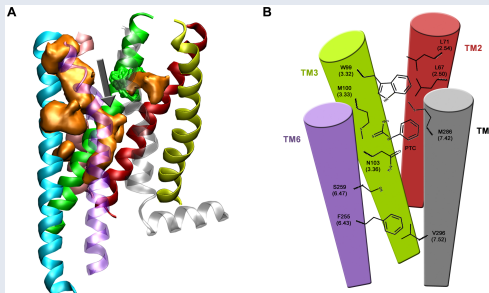


- Signaling cascade: (Chandrashekar et al., 2000; Zhang et al., 2003)
Ligands bind *TAS2R38* → Receptor conformational change → G protein (Gustducin) activation → PLC β 2 → IP₃ → Ca²⁺ release → TRPM5 activation → Depolarization → Neural signal

TAS2R38 SNPs

Molecular Mechanism

- [Bufer et al. \(2005\)](#) stated that **P49A** and **A262V** affect cellular responding most:
 - AVI&AVV variants do not respond to PTC/PROP;
 - PAV&PAI variants respond equally strongly.
- The AVI receptor has a distorted or less accessible binding pocket, leading to poor ligand binding and weak conformational activation. ([Biarnés et al., 2010](#))



TAS2R38 SNPs

Distribution & Evolution

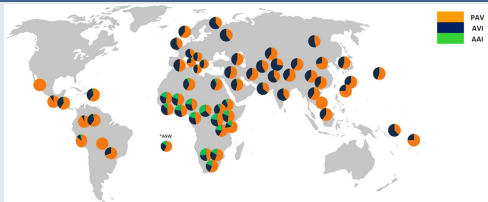


Figure 14: TAS2R38 variant distribution (Risso et al., 2016)

- All 5 samples exhibit wild type, which supports the statistics that PAV variant (wild type) is more prevalent in Asian area.
- Risso et al. (2016) propose “ancient balancing selection acting...before the Out-Of-Africa event, that maintained both PAV and AVI alleles at roughly the same frequency” because they may be “important for detecting potentially toxic bitter compounds found uniquely in the African continent”.

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Thanks!